Reflections

1 Plot point A on a coordinate plane.
Then, plot point A', a reflection of point A across the x-axis.

a $A\,(7, 3)$ b $A\,(-5, -6)$ c $A\,(2, -2)$ d $A\,(-5, 6)$

2 Plot point A on a coordinate plane.
Then, plot point A', a reflection of point A across the y-axis.

a $A\,(6, 9)$. b $A\,(-7, -3)$ c $A\,(9, -10)$ d $A\,(-7, 8)$

3 Consider the following:

a Plot the point $A\,(6, 7)$.

b If point A is reflected about the x-axis to produce point B, determine the coordinates of point B.

c How many units would point A need to be translated to overlap point B?

4 Consider the point $A\,(6, 3)$.

a Plot point A on the number plane.

b If point A is reflected across the y-axis to produce point B, determine the coordinates of point B.

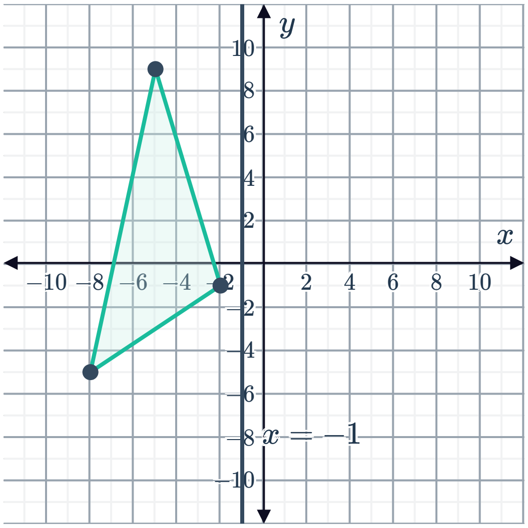
c What distance would point A need to be translated to overlap point B?

5 Plot the line segment AB owhere the endpoints are $A\,(-4, -2)$ and $B\,(6, 7)$. Then plot the reflection of the line segment AB across the x-axis.

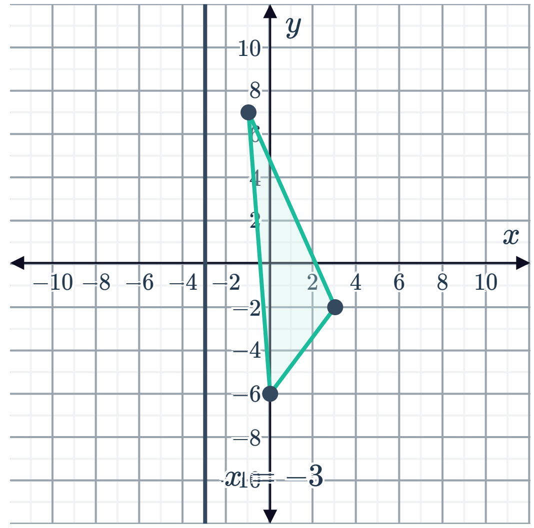
6 Plot the line segment AB, where the endpoints are $A\,(-6, -1)$ and $B\,(10, 8)$. Then, plot the reflection of the line segment about the y-axis.

7 Plot the reflection of each of the following t[illegible]es across the given line

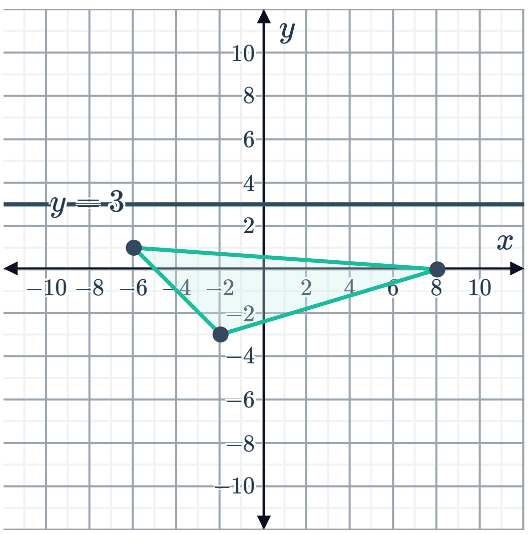
a



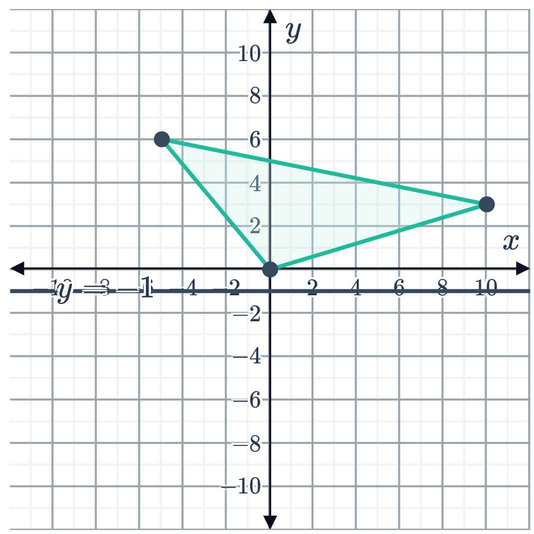
b



c



d

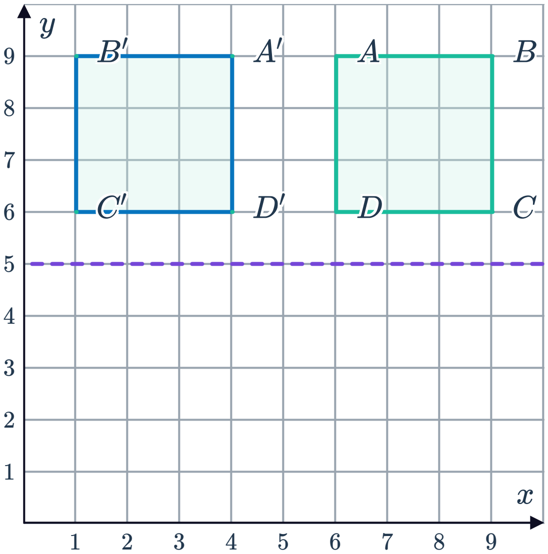


Additional questions

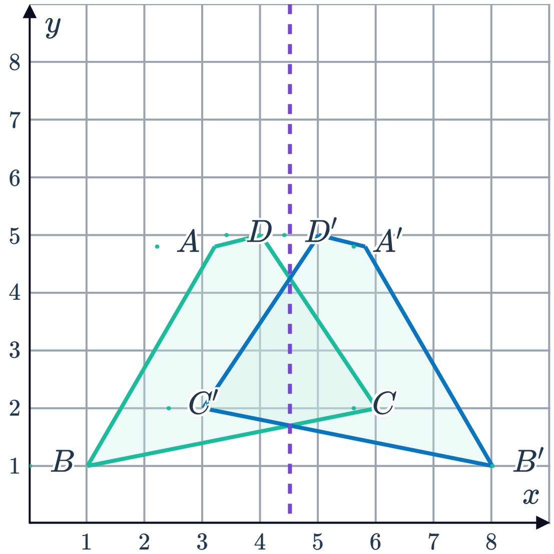
8 Describe the result of reflecting a figure across the x-axis.

9 Determine whether each of the following transformations is a reflection across the given line:

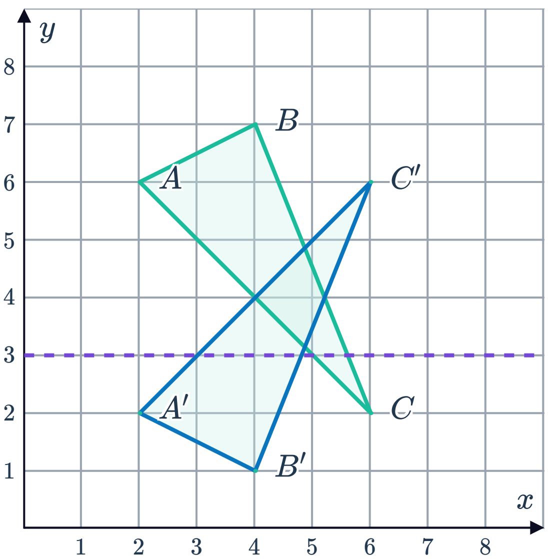
a



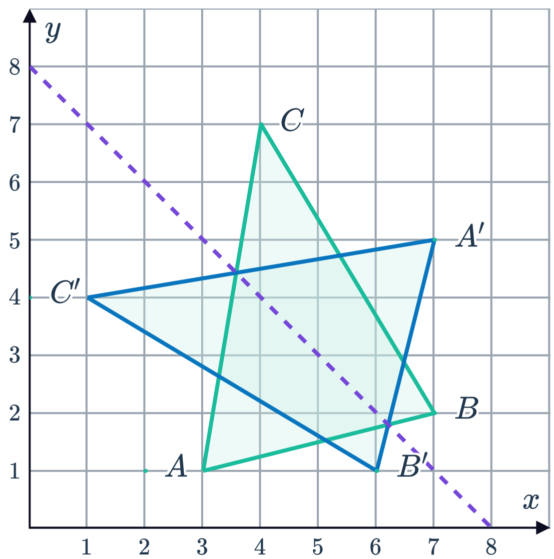
b



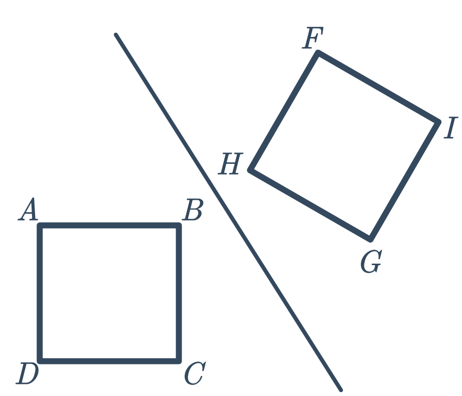
c



d



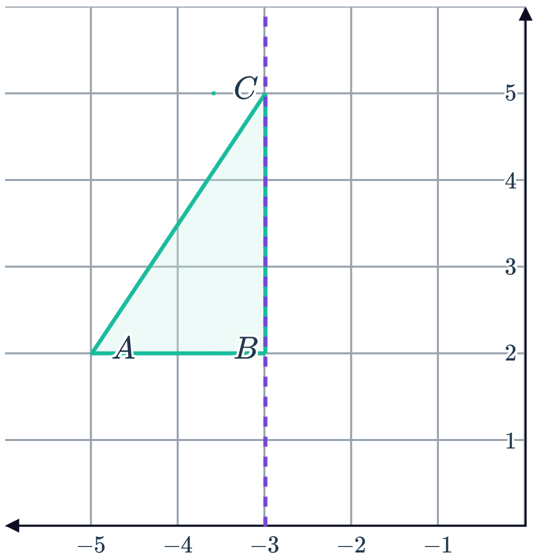
10 Square $ABCD$ was reflected across the given line. Which point in the image corresponds to point A?



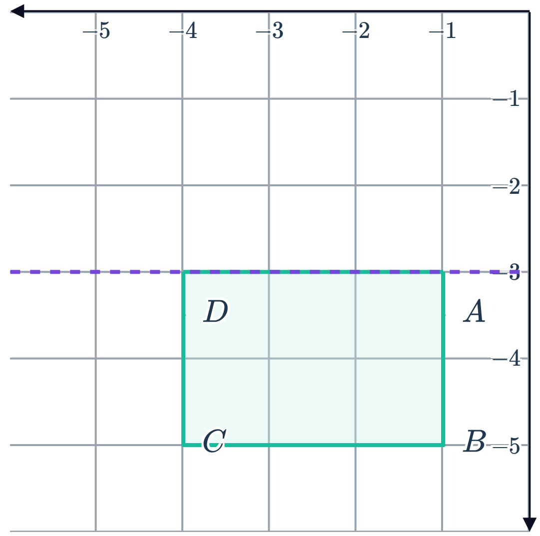
11 Describe the characteristics of a figure that has reflectional symmetry and sketch an example.

12 Reflect each of the following shapes across the given line of reflection. Label all vertices on the image.

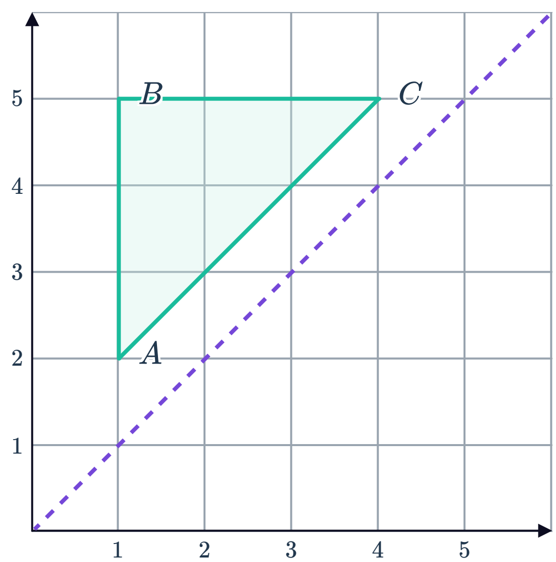
a



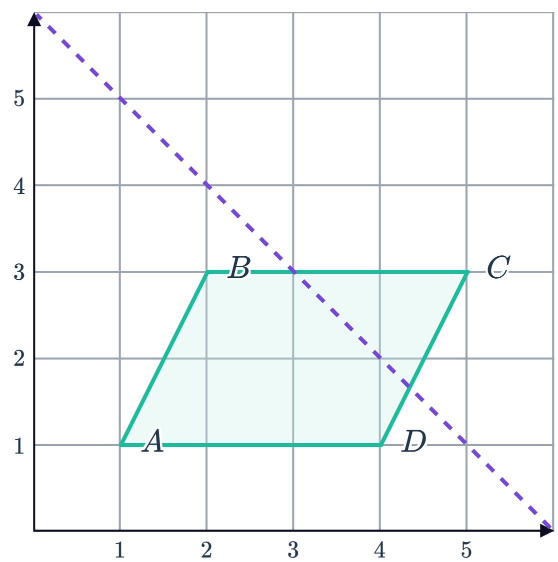
b



c

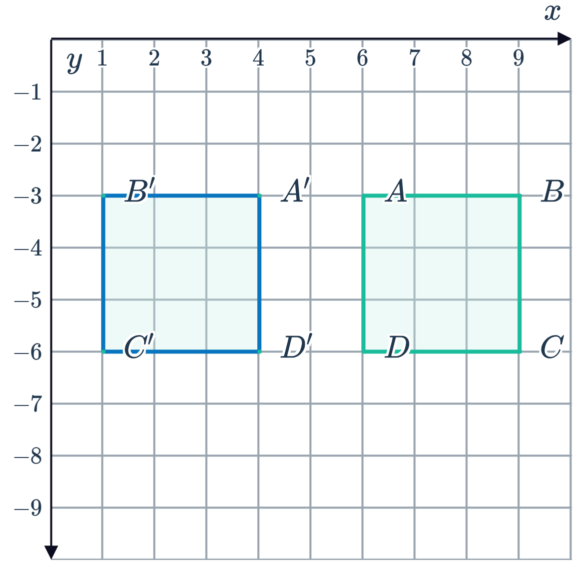


d

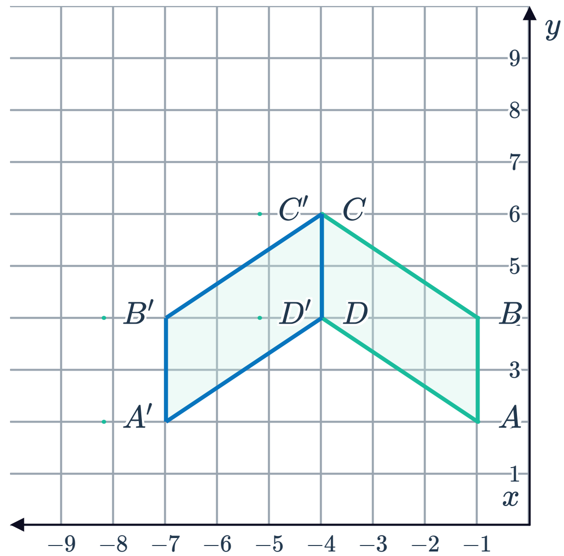


13 Draw the line of reflection for each of the fo[redacted]ng transformations:

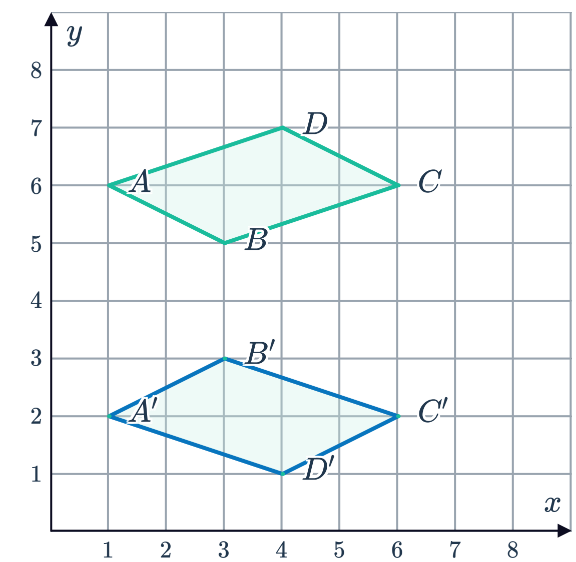
a



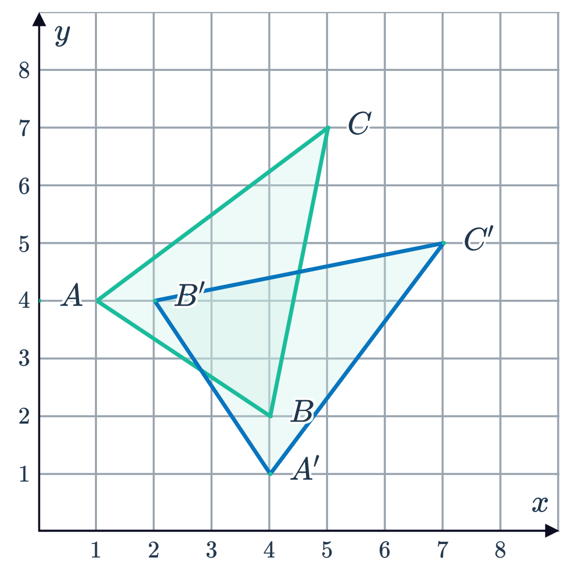
b



c

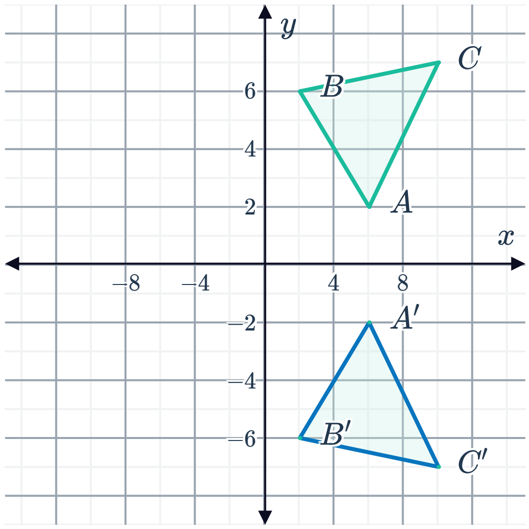


d

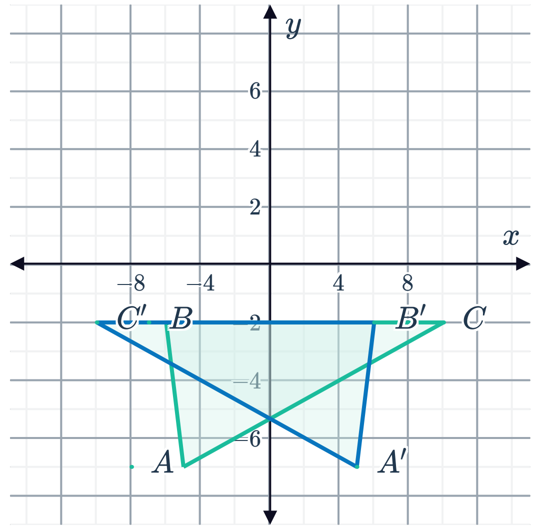


14 Describe each of the following reflections:

a

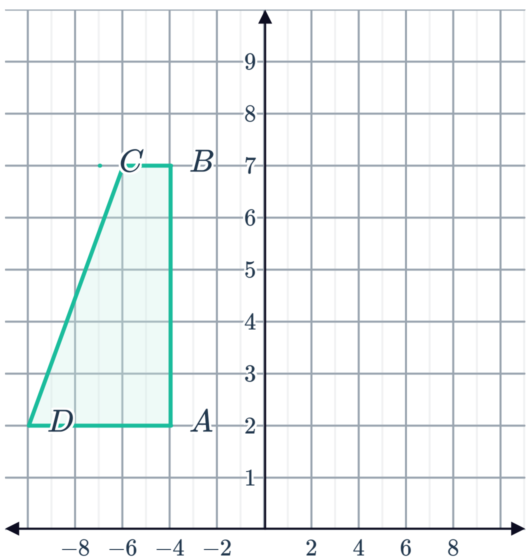


b

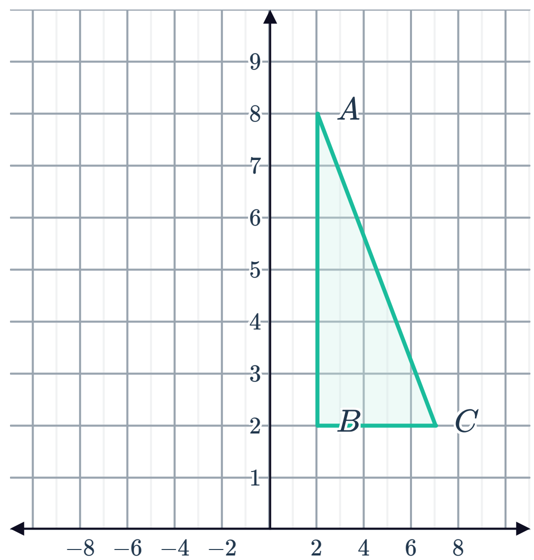


15 Reflect the following shapes across the given the lines of reflection:

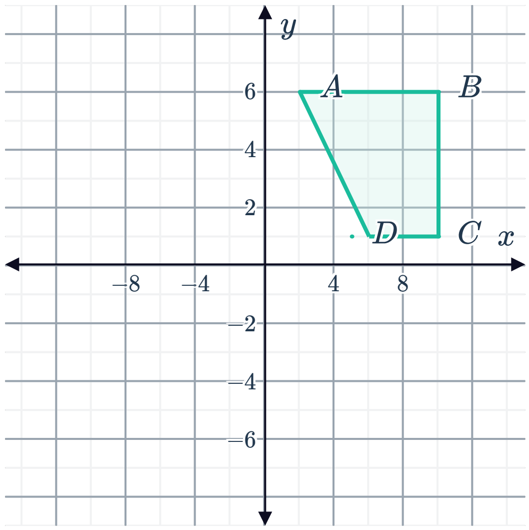
a Line of reflection: y-axis



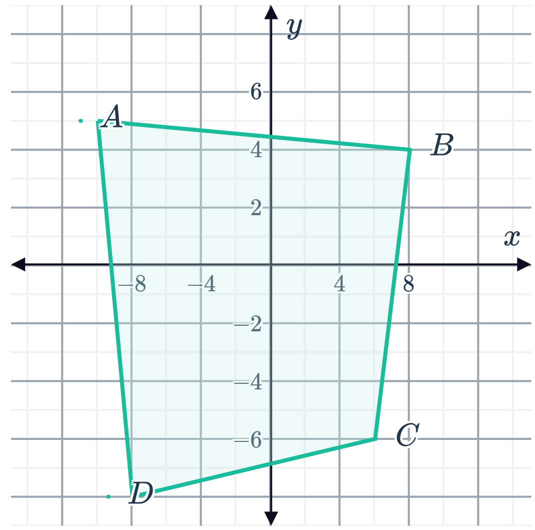
b Line of reflection: y-axis



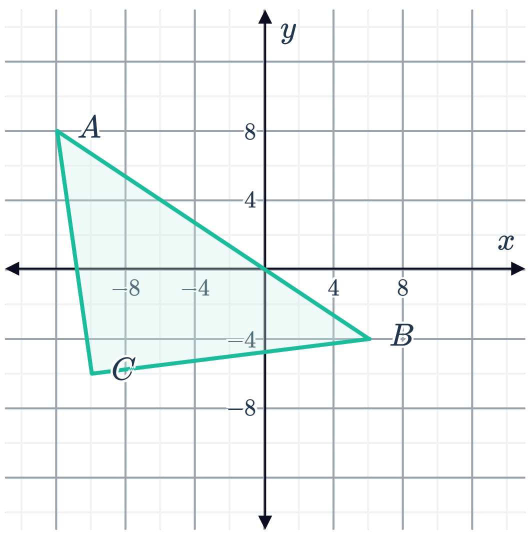
c Line of reflection: x-axis



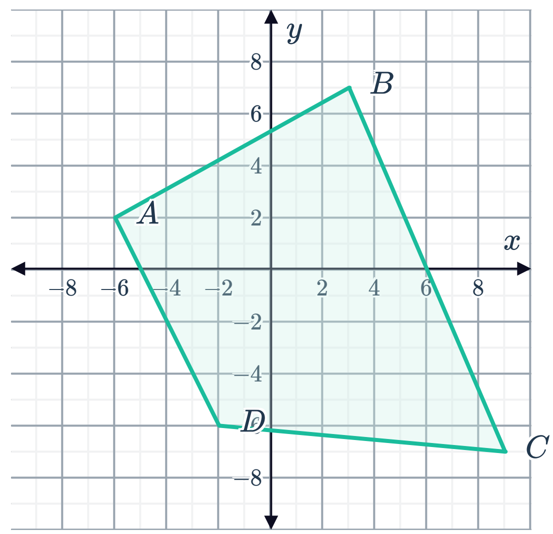
Line of reflection: x-axis



e Line of reflection: $y = x$



f Line of reflection: $y = -x$



16

a If point $M(5, 6)$ is reflected across the y-axis to produce point N, what are the coordinates of point N?

b What distance would the original point M need to be translated to map to point N?

17 If point $F(-3, -7)$ is reflected across the line $y = -x$ to produce point B, what are the coordinates of point B?

18 Determine whether each of the following statements are true or false:

a When a figure is reflected across the x-axis, only the sign of the y-value changes.

b When a figure is reflected across a vertical line, only the y-value changes.

c When a figure is reflected, the preimage and image have the same size and shape.

d When a figure is reflected, the preimage and image have the same orientation.

e A figure is reflected across a line. If the reflected figure is then reflected across the same line again, it will return to the original figure.

19 Sokayna drew figure A and Roly drew figure B.

a Can Sokyana's triangle be reflected to produce Roly's? If so, describe the reflection.

b Roly thinks his triangle has a greater perimeter than Sokanya's. Is this true? Explain why or why not.